Fiscal Year:	FY 2005	**Task Last Updated:**	FY 10/20/2004
PI Name:	Marston, Philip L. Ph.D.		
Project Title:	Acoustic Radiation Pressure Control of Confined and Open Liquid Interfaces		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Complex fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	marston@wsu.edu	**Fax:**	FY 509-335-7816
PI Organization Type:	UNIVERSITY	**Phone:**	509-335-5343
Organization Name:	Washington State University		
PI Address 1:	Department of Physics		
PI Address 2:			
PI Web Page:			
City:	Pullman	**State:**	WA
Zip Code:	99164-2814	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	01/08/2004	**End Date:**	09/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Niederhaus, Charles	**Contact Phone:**	(216) 433-5461
Contact Email:	charles.niederhaus@grc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Thiessen, David B. (Washington State University)		
Grant/Contract No.:	NNC04GA53G		
Performance Goal No.:			
Performance Goal Text:			
	This proposal is to investigate the use of acoustic radiation pressure to control and manipulate gas-liquid systems relevant to two-phase flow and interfacial stability in reduced gravity. In previous research [Marr-Lyon, Thiessen, & Marston, "Passive stabilization of capillary bridges in air with acoustic radiation pressure," Phys. Rev. Lett. 86, 2293-2296 (2001); Marr-Lyon, Thiessen, & P. L. Marston, "Stabilization of a cylindrical capillary bridge far beyond the Rayleigh-Plateau limit using acoustic radiation pressure and active feedback," J. Fluid Mech. 351, 345-357 (1997)] we have demonstrated that long liquid bridges may be stabilized by the application of suitable ultrasonic fields. This proposal will build upon that background to examine the ultrasonic response and/or control of other two-phase situations relevant to microgravity, including: (1) liquid lenses (slugs), bubbles, and collars confined to tubes; (2) stability of low-speed liquid jets and controlled breakup into drops; (3) active acoustic damping of capillary modes; and (4) liquid coatings (internal or external) on cylinders. Much of the research concerns the radiation pressure of ultrasonic guided		

Task Description:	waves and in some of the jet experiments, the wave is internal to the jet (with a frequency close to or above an acoustic waveguide cut-off frequency). Some related ultrasonic aspects of the liquid bridge research may continue (depending on the results of new experiments in normal gravity) if the KC-135 remains available. The liquid jet experiments and some of the other experiments may also be appropriate for the KC-135. The primary objective is to determine if guided sound may be used to significantly affect the stability and flow of liquids and gasses in two-phase contact. In some cases it will be desirable to alter the stability of low speed liquid jets using ultrasound. In others it will be desirable to use the radiation pressure of sound to clear or move liquid slugs in a channel or confined passage. It is also plausible that acoustic radiation pressure and streaming may be used to enhance mixing and heat or mass transfer in capillary pipes. The radiation pressure of sound in a gas may also be used to control the thickness and stability of liquid films on a solid surface. The ability to control the dynamics of interfaces in the aforementioned situations will enhance the reliability of two-phase systems in reduced gravity. If acoustic radiation pressure and streaming can be used to enhance the mixing and heat or mass transfer from liquid columns in the form of capillary pipes, the size of certain two-phase exchange systems could be reduced. It is anticipated there will be applications to life support systems. It is plausible that there will also be medical applications to the clearing of blockages in air passageways. The use of high frequency sound as a means of transporting momentum has some potential advantages when compared to light and other electromagnetic waves. The extremely small value of the speed of sound compared to the speed of light means that the required ultrasound intensity is much smaller than the optical intensity needed to generate a given radiation stress.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	While especially suitable for enhanced performance in reduced gravity two-phase systems, applications in normal grants are anticipated.
Task Progress:	This project investigates the ultrasonic control of liquids in new reduced gravity two-phase situations with potential applications to stabilization and mixing. It is plausible that the ultrasonic methods we have explored thus far may have their most significant application in reduced gravity in the enhanced mixing and heat and mass transfer from capillary pipes resulting from acoustic radiation pressure and streaming. The emphasis during FY2004 has been on experimentally exploring the responses of liquid jets and liquid columns to modulated acoustic radiation pressure in the case where the ultrasound is guided by the liquid. In theoretical work we have identified an appropriate superposition of ultrasonic waves in the liquid that has the desirable property of having a spatial as well as temporal modulation. We also identified configurations with a pure axial acoustic mode that lack a significant spatial variation of the acoustic radiation pressure. As an aid to understanding the propagation down the tapered jets and columns present in normal gravity we have applied finite element methods (FEM). Those calculations reveal the importance of taper on affecting the acoustic cut-off frequency of the waveguide. Other areas of theoretical progress include: (1) We developed approximations for the acoustic radiation damping for the ultrasound in an ideal liquid jet. These calculations reveal that the radiation damping is usually negligible except near waveguide cut-off. (2) We analyzed the complex dispersion relations for capillary waves on jets and cylindrical columns. These calculations have been helpful for anticipating the response to space-time modulated radiation pressure. The experimental progress was aided by our procurement of a high-speed video system for monitoring liquid responses at 3000 frames per second. We developed transducers for the efficient coupling of sound into a liquid jet or column waveguide. We have used these to demonstrate the ultrasonic excitation of the Rayleigh capillary instability of a liquid jet. We also developed and used a light scattering method for monitoring the response of a jet. Using this method we confirmed that the response of the jet to modulated internal acoustic radiation pressure is proportional to the square of the ultrasonic amplitude [J. B. Lonzaga, C. F. Osterhourdt, D. B. Thiessen, and P. L. Marston, (abstract) J. Acoust. Soc. Am. (to be published in 2004)]. By placing transducers on both ends of a vertical liquid column we have also measured the ultrasonic cut-off frequency for the propagation of sound down the liquid waveguide. As a spin-off of this study, our investigation of ultrasonic propagation down jets will be helpful in improving the understanding of using liquid jets in ultrasonic non-destructive testing in the aerospace industry. Thus far, that method appears to be treated as an art without a significant development of the underlying theory. A second aspect of the research concerns the application of ultrasound in air to affect the dynamics of liquid slugs and blockages within a narrow tube. While we have demonstrated an ultrasonic coupling to the motion of slugs, that research was not emphasized in the nine month effort summarized here.
Bibliography Type:	Description: (Last Updated: 06/21/2021)
Abstracts for Journals and Proceedings	Marston PL. "Shape oscillations of acoustically levitated drops in water: Early research with Bob Apfel on modulated radiation pressure." Invited talk. Acoustical Society of America meeting, May 2004. J Acoust Soc Am. 2004;115:2419. , May-2004
Abstracts for Journals and Proceedings	Thiessen DB, Lonzaga JB, Osterhoudt CF, Wei W, Marston PL. "Manipulation of fluid objects using acoustic radiation pressure: Beyond drops and bubbles and retrospection." Invited talk: 148th Meeting of the Acoustical Society of America. San Diego, California, USA, 15-19 November 2004. J Acoust Soc Am. 2004;116:2597(A). , Nov-2004
Abstracts for Journals and Proceedings	Lonzaga JB, Osterhoudt CF, Thiessen DB, Marston PL. "Optical detection of the response of liquid jets to internal modulated ultrasonic radiation pressure." 148th Meeting of the Acoustical Society of America. San Diego, California, USA, 15-19 November 2004. J Acoust Soc Am. 2004;116:2598(A). , Nov-2004
Abstracts for Journals and Proceedings	Marston PL, Wei WW, Thiessen DB. "Acoustic radiation force on elliptical cylinders and spheroidal objects in low frequency standing waves." 17th International Symposium on Nonlinear Acoustics, June 2005. Program of the 17th International Symposium on Nonlinear Acoustics, June 2005. , Jun-2005
Articles in Peer-reviewed Journals	Wei W, Thiessen DB, Marston PL. "Control of the natural frequency of the (2,0) mode of liquid bridges using active electrostatic fields." Physics of Fluids. 2005 Jul;17(7):071702. https://doi.org/10.1063/1.1978947 , Jul-2005
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Awards	Marston PL. "Philip L. Marston (Washington State University) was awarded the Silver Medal in Physical Acoustics by the Acoustical Society of America (November 2003)." Nov-2003
Dissertations and Theses	Wei W. "Capillary Bridge Stability and Dynamics: Active Electrostatic Stress Control and Acoustic Radiation Pressure." Ph.D. thesis, Washington State University (May 2005). , May-2005